# DHAIRYA BHATIA

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## RESEARCH INTERESTS

Video understanding, embodied intelligence, egocentric and exocentric video, evaluation of video-language models, and multimodal agents.

## EDUCATION

**Khoury College of Computer Sciences, Northeastern University** — Boston, MA
*Master of Science in Computer Science* — Sep 2025 – May 2027 (Expected)

- Research advisor: Prof. Sarah Ostadabbas, Augmented Cognition Lab (ACLab).

**The University of British Columbia** — Kelowna, BC
*Bachelor of Science in Computer Science* — Sep 2020 – May 2024

## PUBLICATIONS

**Dhairya Bhatia**, Bishoy Galoaa, et al. *MotionBlind: Probing the Illusion of Motion Understanding in Video-LLMs*. arXiv:2609.09528, 2026.

## RESEARCH EXPERIENCE

**Augmented Cognition Lab (ACLab), Northeastern University** — May 2026 – Present
*Research Assistant · Advisor: Prof. Sarah Ostadabbas*

***MotionBlind Benchmark***

- Co-led MotionBlind, a contrastive benchmark of 60 self-recorded video pairs that share the same person, scene, and camera but differ only in speed, magnitude, or direction.
- Built SLURM evaluation pipelines on Northeastern's HPC cluster to benchmark 8 Video-LLMs on MotionBlind, TimeBlind, and Video-MME, showing open models score 52 to 59% on single questions but at most 11.7% on full pairs against 91.3% for humans.
- Ran frame-shuffling, frame-removal, and frame-selection studies across 1 to 24 frame budgets, showing open models drop to near 0% without ordered video and that no selector beats uniform sampling.

***CV4Smalls 2026 CVPR Challenge Dataset***

- Managed the JSONL labeling pipeline for the CV4Smalls 2026 CVPR challenge dataset of 108 video pairs across 9 categories, fixing swapped videos and mismatched question pairings.

***Egocentric and Exocentric Video Dataset (in progress)***

- Collecting an egocentric and exocentric video dataset.

**CACTILab, Northeastern University** — Mar 2026 – Present
*Research Volunteer*

***C-to-Rust Translation***

- Evaluated multiple C-to-Rust translation tools.

***Multi-Agent Firmware Generation***

- Built a multi-agent LLM pipeline that writes embedded firmware and feeds compiler errors back to the agents, with model tiering and per-turn cost tracking.
- Delivered a working firmware MVP generated end to end by the pipeline and tested on a TI MSPM0L2228 board.

## TEACHING

**Teaching Assistant, CS3100, Northeastern University** — Jan 2026 – Jun 2026

- Supported students in labs and office hours and helped design and test labs and assignments before release.
- Led analysis of student feedback on AI-assisted programming workflows, informing curriculum design.

## INDUSTRY EXPERIENCE

**Oriv · Software Development Intern / AI Developer** — Sep 2024 – Jul 2025

- Prototyped knowledge-graph methods for fitting graph structures onto IoT sensor signals for industrial automation POCs.

- Introduced sensor-virtualization methods enabling data simulation and reducing dependency on physical hardware.

## SELECTED PROJECTS

**Project Escalion · Rust Game Engine**

- Building a custom Rust game engine with a wgpu renderer and a sparse-set entity component system for high-performance real-time rendering.

**MITRE eCTF 2026 · Embedded Security Competition**

- Contributed firmware implementation, code review, and bug fixes in a security-critical, hardware-constrained environment.

## LEADERSHIP AND SERVICE

**Co-Founder and Co-President, Project Escalion** Northeastern University

- Co-founded a Rust game development studio that grew into a Khoury-recognized graduate club with game design, engine development, and multiplayer tracks.

**Senator for Student Affairs, Graduate Student Government** Northeastern University

- Named GSG Senator of the Month in February 2026 while representing graduate students on student affairs.

## TECHNICAL SKILLS

**ML / AI:** PyTorch, Multimodal Learning, VLMs, Video-LLM Evaluation and Benchmarking, Computer Vision
**Languages:** Python, Rust, C, Java, SQL, Bash
**Systems and Tools:** Linux, SLURM, HPC, Docker, Git, Embedded Systems